

CSE 2215: Data Structure and Algorithms-I

- 1
 - a) Distinguish between Array and Linked list.
 - b) What are the differences between static and dynamic memory allocation?
- 2
 - a) Draw diagrams for a linear, doubly and circular linked list
 - b) Show memory mapping for each of the diagrams of Ques 2(a).
- 3 Show variable declarations for each of the structure type pointer variables given below
 - a) student (name, id, marks, next)
 - b) head(data, next)
 - c) super(back, data, next)
 - d) start(row, col, data, right, down)
 - e) root(left, info, right)

Where name is string type

id is int type

marks is float type

info is char type

row, col, data are int type

next represents address of next node

back represents address of previous node

right represents address of right node

down represents address of down node

left represents address of left node

- 4 Suppose a linear linked list headed with “start” contains four nodes whose data values are 10, 20, 30, 40, respectively. Show the following operations
 - i) Draw a diagram for the linked list
 - ii) What are the names of the nodes that contain 10, 20, 30, 40, respectively? Show the names of the nodes in the linked list diagram
 - iii) Write statements to represent 10, 20, 30, 40, respectively
 - iv) Write a statement to represent NULL at the end of the linked list

- 5 Show the effects of each of the following statements given using the following structure. Assume, x=LAST_DIGIT_OF_YOUR_STUDENT_ID.

```
struct list {  
    int data;  
    struct list *next;  
};  
typedef struct list node;  
node *super, *temp, *temp1;
```

Statements:

```
super=(node*)malloc(sizeof(node));  
super->next=super;  
super->data=x*3;  
temp=(node*)malloc(sizeof(node));
```

```

temp->data=super->data+ x*5;
temp->next=super;
temp1=(node*)malloc(sizeof(node));
temp1->data=super->data * temp->data;
temp1->next=super;
super->next=temp1;
super->next->next=temp;

```

6 Write algorithms for the followings using the structure given below

```

struct list {
    int data;
    struct list *next;
};
typedef struct list node;

```

- a) To create a linear linked list in memory
 - b) To display the contents of a linear linked list
 - c) To insert an element in a linear linked list
 - d) To delete an element from a linear linked list
 - e) To concatenate two linear linked list
 - f) To reverse a linear linked list
 - g) To count the length of a linear linear linked list
 - h) To calculate the sum of all the data in the linked list
- 7
- a) What are merits and demerits of doubly linked list over linear linked list?
 - b) Distinguish between linear and circular linked lists.
- 8
- a) Implement all the Question of Ques (6) using double linked link list
 - b) Implement logic to create a doubly linked list using all the individual digits of your own STUDENT ID. Use the following structure and variables to design the logic.

```

struct list{
    int data;
    struct list *next;
    struct list *back;
};
typedef struct list node;
node *start, *temp;
For example, STUDENT ID=011202074
Individual digits are 0, 1, 2, 4, 7
Linked List is: NULL←0↔1↔2↔4↔7→NULL
Assume starting node is “start”

```

- 9 Write algorithms using a linear circular linked list for
- i) First insertion
 - ii) First deletion
 - iii) Last insertion
 - iv) Last deletion